

Effective thermal conductivity from a phase field: a diffuse cell problem without first-order interface bias

Abstract

We compute the effective thermal conductivity $\mathbf{K}_{\text{eff}}$ of evolving snow microstructures by solving the steady-state homogenization cell problem of Calonne et al. (2011) directly on the phase field, where the ice–air interface is a smooth tanh transition of width a few ε rather than a sharp surface. The conductivity must be interpolated across that transition, and a matched asymptotic analysis shows that, to first order in ε , the transition acts like a sharp interface with a spurious resistance to heat crossing it and a spurious conductance along it; the standard arithmetic law removes the second but not the first, because resistivity $1/K$ is convex in ϕ . At $K_i/K_a \approx 115$ the effect is large: for a planar ice slab with heat flowing across it at $\varepsilon = L/50$, the computed conductivity is $0.063 \text{ W m}^{-1}\text{K}^{-1}$ against an exact 0.040 for the same geometry, 59% too high, and because the error scales with ε times the interface area per unit volume, which sintering reduces, it distorts the time trend of $\mathbf{K}_{\text{eff}}$ as well as its level. We remove both terms by interpolating the conductivity arithmetically for gradients along the interface, where the sharp solution’s temperature gradient is continuous, and harmonically for gradients across it, where the heat flux is continuous; each choice makes its spurious term vanish exactly for the tanh profile, the interface normal comes from $\nabla\phi$, and the law reduces to the pure-phase values outside the transition, so nothing is fitted and the sharp limit is unchanged. Against closed-form sharp solutions it reproduces a planar slab to 4×10^{-7} at every width tested (the arithmetic law is 3.8–59% high) and a disk array to 0.006% at $\varepsilon = L/800$ with the error falling faster than second order, while the arithmetic law’s error matches our first-order prediction to within 2%.

1 The sharp cell problem

A periodic cell Ω is split into ice Ω_i and air Ω_a , which meet on the interface Γ . The conductivities $K_i, K_a > 0$ are constant. Let $\hat{\mathbf{n}}$ be the unit normal on Γ pointing into the ice, and let $\llbracket f \rrbracket = f|_i - f|_a$ denote the jump across Γ . Following Calonne et al. (2011), $\mathbf{K}_{\text{eff}}$ is obtained from two cell-periodic correctors, $\mathbf{t}_i$ defined on the ice only and $\mathbf{t}_a$ on the air only:

$$\nabla \cdot (K_i(\nabla \mathbf{t}_i + \mathbf{I})) = \mathbf{0} \text{ in } \Omega_i, \quad \nabla \cdot (K_a(\nabla \mathbf{t}_a + \mathbf{I})) = \mathbf{0} \text{ in } \Omega_a, \quad (1)$$

$$\mathbf{t}_i = \mathbf{t}_a \text{ on } \Gamma, \quad K_i(\nabla \mathbf{t}_i + \mathbf{I})\hat{\mathbf{n}} = K_a(\nabla \mathbf{t}_a + \mathbf{I})\hat{\mathbf{n}} \text{ on } \Gamma, \quad (2)$$

normalized by $\int_{\Omega_i} \mathbf{t}_i + \int_{\Omega_a} \mathbf{t}_a = \mathbf{0}$, with

$$\mathbf{K}_{\text{eff}} = \frac{1}{|\Omega|} \left[\int_{\Omega_i} K_i(\nabla \mathbf{t}_i + \mathbf{I}) + \int_{\Omega_a} K_a(\nabla \mathbf{t}_a + \mathbf{I}) \right]. \quad (3)$$

Column m of this problem is the temperature field $T = \mathbf{e}_m \cdot \mathbf{x} + t_m$ driven by a unit mean gradient along $\mathbf{e}_m$. The two interface conditions are the physics: temperature is continuous, and so is the normal heat flux. Below we write $\mathbf{q} = \mathbf{K}\nabla T$ for the flux, a sign convention that is standard for the cell problem (the physical heat flux is $-\mathbf{q}$).

A useful consequence of (2): on Γ the *tangential* temperature gradient $\mathbf{E}_t = (\mathbf{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}})\nabla T$ is continuous, because T is, and the *normal flux* $q_n = \mathbf{q} \cdot \hat{\mathbf{n}}$ is continuous. The normal gradient and the tangential flux jump, by the factor $K_i/K_a \approx 115$.

2 One field on one domain

Let χ be the indicator of the ice ($\chi = 1$ in Ω_i , 0 in Ω_a), and define

$$K_\star = K_i\chi + K_a(1 - \chi), \quad \mathbf{t} = \mathbf{t}_i\chi + \mathbf{t}_a(1 - \chi). \quad (4)$$

The two correctors form one H^1 field. For a smooth periodic vector field ψ , integrating by parts on each phase gives

$$\int_{\Omega} \mathbf{t} \cdot \nabla \cdot \psi = - \int_{\Omega_i} \nabla \mathbf{t}_i \cdot \psi - \int_{\Omega_a} \nabla \mathbf{t}_a \cdot \psi - \int_{\Gamma} \llbracket \mathbf{t} \rrbracket \psi \cdot \hat{\mathbf{n}} d\Gamma.$$

The last term vanishes for every ψ if and only if $\mathbf{t}_i = \mathbf{t}_a$ on Γ . So, by the first condition in (2), $\mathbf{t} \in H_{\text{per}}^1(\Omega)$, and its gradient is the piecewise gradient, with no part concentrated on Γ . The coefficient $K_{\star}$ may jump; the corrector may not.

One weak equation. Multiply each bulk equation by a periodic test function v , integrate by parts on its own phase, and add. The outer-boundary terms cancel by periodicity. The interface terms combine into $\int_{\Gamma} v \llbracket K(\nabla \mathbf{t} + \mathbf{I}) \hat{\mathbf{n}} \rrbracket$, which vanishes by the second condition in (2). The two volume integrals have the same integrand once written with $K_{\star}$ and the glued $\mathbf{t}$, so they merge:

$$\int_{\Omega} \nabla v \cdot K_{\star}(\nabla \mathbf{t} + \mathbf{I}) d\Omega = 0 \quad \forall v \in H_{\text{per}}^1(\Omega), \quad \langle \mathbf{t} \rangle = \mathbf{0}, \quad \mathbf{K}_{\text{eff}} = \langle K_{\star}(\nabla \mathbf{t} + \mathbf{I}) \rangle. \quad (5)$$

Both steps reverse. A test function supported inside one phase returns that phase's bulk equation, and one straddling Γ returns the flux condition. A single H^1 field cannot jump, so continuity is built in. So (5) is the Calonne problem, written on one domain with one unknown, and not an approximation to it.

3 The diffuse formulation

The phase-field model does not provide χ . It provides a smooth ϕ , equal to 1 in ice and 0 in air, with the tanh profile

$$\phi = \ell(s/\varepsilon) = \frac{1}{2} [1 + \tanh(s/2\varepsilon)], \quad \ell(u) = \frac{1}{1 + e^{-u}}, \quad \ell(-u) = 1 - \ell(u), \quad (6)$$

where s is the signed distance to Γ (the $\phi = \frac{1}{2}$ level set), positive in ice. ℓ is simply this profile written in the variable $u = s/\varepsilon$.

Lemma 1. *Let $K(\cdot)$ be any function with $K(0) = K_a$ and $K(1) = K_i$. Then $K(\chi) = K_{\star}$ exactly.*

Proof. χ only takes the values 0 and 1. □

So (5) is the equation $\int \nabla v \cdot K(\chi)(\nabla \mathbf{t} + \mathbf{I}) = 0$, and the diffuse formulation is the same equation with χ replaced by ϕ :

$$\int_{\Omega} \nabla v \cdot \mathbf{K}(\phi)(\nabla \mathbf{t} + \mathbf{I}) d\Omega = 0 \quad \forall v \in H_{\text{per}}^1(\Omega), \quad \mathbf{K}_{\text{eff}}^{\varepsilon} = \langle \mathbf{K}(\phi)(\nabla \mathbf{t} + \mathbf{I}) \rangle. \quad (7)$$

The only approximation is geometric, ϕ in place of χ . The lemma also shows that the sharp problem cannot see how $\mathbf{K}$ behaves for $0 < \phi < 1$. Every interpolation with the right end values gives the same sharp limit. The choice matters only inside the transition, and there it sets the error at finite ε . The standard choice is the arithmetic law $K(\phi) = K_a + (K_i - K_a)\phi$. The question is which interpolation makes (7) closest to (5).

To answer it we allow, from the start, a coefficient that may act differently along and across the interface:

$$\mathbf{K} = K_t(\phi)(\mathbf{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}) + K_n(\phi)\hat{\mathbf{n}} \otimes \hat{\mathbf{n}}, \quad \hat{\mathbf{n}} = \frac{\nabla \phi}{|\nabla \phi|}, \quad (8)$$

with K_t, K_n equal to K_a at $\phi = 0$ and K_i at $\phi = 1$. A scalar law is the special case $K_t = K_n$.

Where this form comes from. The outer product is a projection: $(\hat{\mathbf{n}} \otimes \hat{\mathbf{n}})\mathbf{g} = \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{g})$ keeps the part of a vector $\mathbf{g}$ along $\hat{\mathbf{n}}$, and $\mathbf{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}$ keeps the part along the interface. So (8) says only that the normal part of the gradient is multiplied by K_n and the tangential part by K_t . It is also the *only* linear law with that property and no coupling between the two directions. Suppose $\mathbf{K}$ had a normal–tangential entry c . Inside the transition the normal flux would be $K_n \partial_s T + c \mathbf{E}_t$, and the expansion of §4 would give the temperature jump an extra term $-\mathbf{E}_t \int c/K_n du$, a spurious interface term beyond the two analysed below. With $c = 0$, $\mathbf{K}$ is symmetric, $\hat{\mathbf{n}}$ is an eigenvector with eigenvalue K_n , and the directions along the interface are eigenvectors with eigenvalue K_t . A symmetric tensor with that eigenstructure is exactly (8). In two dimensions, with $\hat{\mathbf{n}} = (\cos \theta, \sin \theta)$,

$$\mathbf{K} = \begin{pmatrix} K_t \sin^2 \theta + K_n \cos^2 \theta & (K_n - K_t) \sin \theta \cos \theta \\ (K_n - K_t) \sin \theta \cos \theta & K_t \cos^2 \theta + K_n \sin^2 \theta \end{pmatrix}.$$

Its off-diagonal entries in x, y are not a coupling. They appear only because the interface is not aligned with the axes, and in the interface’s own frame $\mathbf{K} = \text{diag}(K_n, K_t)$. For the profile (6), $\nabla \phi = \ell'(s/\varepsilon) \nabla s/\varepsilon$ is parallel to ∇s , so $\hat{\mathbf{n}}$ is exactly the normal to the level sets of ϕ , and the phase field supplies it without any extra construction.

4 What the transition does to the solution

We analyse (7) by matched asymptotic expansions in ε . Away from Γ (the *outer* region), ϕ is 0 or 1 up to exponentially small terms, and the temperature satisfies the pure-phase equations. Near Γ (the *inner* region), use the signed distance s and coordinates $\mathbf{y}$ along Γ , and stretch $s = \varepsilon u$. The coefficient (8) depends on u only, through $\ell(u)$. In these coordinates $\nabla \cdot (\mathbf{K} \nabla T) = 0$ reads

$$\varepsilon^{-2} \partial_u (K_n \partial_u T) + \varepsilon^{-1} \kappa K_n \partial_u T + \nabla_{\mathbf{y}} \cdot (K_t \nabla_{\mathbf{y}} T) + O(\varepsilon) = 0, \quad (9)$$

with κ the curvature of Γ (the sum of its principal curvatures). We assume the transition is thin compared with the radius of curvature and with the distance to other parts of the interface. Expand $T = T_0 + \varepsilon T_1 + \varepsilon^2 T_2 + \dots$ in the inner region.

Order ε^{-2} : temperature is continuous. $\partial_u (K_n \partial_u T_0) = 0$, so $K_n \partial_u T_0$ is constant in u . Matching to a bounded outer field forces that constant to be zero, so T_0 does not depend on u . At leading order the temperature, and therefore its tangential gradient $\mathbf{E}_t = \nabla_{\mathbf{y}} T_0$, is the same at every point across the transition.

Order ε^{-1} : normal flux is continuous. $\partial_u (K_n \partial_u T_1) = 0$ (the curvature term multiplies $\partial_u T_0 = 0$), so

$$K_n(\ell(u)) \partial_u T_1 = q_n(\mathbf{y}), \quad \partial_u T_1 = \frac{q_n}{K_n(\ell(u))}, \quad (10)$$

with q_n the same at every point across the transition, and equal to the outer normal flux on both sides. Inside the transition the normal gradient varies as $1/K_n$ while the flux stays fixed.

The equation itself thus reproduces the structure of the sharp solution: the tangential gradient and the normal flux do not change across the transition, exactly the quantities the sharp interface conditions keep continuous. What the transition can alter is how much temperature drop it takes to carry q_n across it, and how much flux it carries along it for a given $\mathbf{E}_t$.

First order: the effective interface conditions. Integrating (10) across the transition gives the total temperature change. The outer solution, extrapolated to Γ from each side, has slopes q_n/K_i and q_n/K_a . Matching the two at the edges $u = \pm U$ of the inner region and letting $U \rightarrow \infty$ gives the outer temperature jump

$$[[T]] = \Sigma_n q_n, \quad \Sigma_n = \varepsilon \int_{-\infty}^{\infty} \left[\frac{1}{K_n(\ell(u))} - \frac{1}{K_n(H(u))} \right] du, \quad (11)$$

where H is the unit step, so $K_n(H(u))$ is the sharp conductivity $K_\star$ seen along the normal. Integrating the $O(1)$ equation of (9) across the transition in the same way gives the outer normal-flux jump

$$[[q_n]] = -\nabla_{\Gamma} \cdot (\Sigma_t \nabla_{\Gamma} T), \quad \Sigma_t = \varepsilon \int_{-\infty}^{\infty} [K_t(\ell(u)) - K_t(H(u))] du. \quad (12)$$

The curvature term contributes κq_n to both the inner and the extrapolated outer fields, so it cancels from the jump at this order. The integrals converge because $\ell - H$ decays exponentially.

Proposition 1. *To first order in ε , the diffuse problem (7) is the sharp problem with its interface conditions (2) replaced by (11)–(12). Σ_n acts as a spurious thermal resistance across Γ , and Σ_t as a spurious conductance along it. Consequently, the diffuse formulation reproduces the sharp one to first order, for any microstructure, if and only if*

$$\Sigma_t = 0 \quad \text{and} \quad \Sigma_n = 0. \quad (13)$$

If both vanish, the $O(\varepsilon)$ correction to the outer field solves the sharp problem with zero data, so it is zero, and $\mathbf{K}_{\text{eff}}^\varepsilon - \mathbf{K}_{\text{eff}}^\star = O(\varepsilon^2)$. The two numbers depend only on the profile and on K_t, K_n . They are local properties of the transition, not of the microstructure, so (13) is a condition on the interpolation alone.

5 The arithmetic law fails the normal condition

Step 1: each excess is ε times a number. The integrals in (11)–(12) depend only on K_i, K_a and the interpolation, not on ε or the geometry. Unless one is zero, the corresponding error is first order in ε .

Step 2: a quantity affine in ϕ has zero excess. If $f(\phi) = \alpha + \beta\phi$, then $f(\ell(u)) - f(H(u)) = \beta[\ell(u) - H(u)]$. By the antisymmetry in (6), $\ell - H$ is odd: for $u > 0$, $\ell(u) - 1 = -\ell(-u)$. Its integral is zero (Fig. 1). The transition adds exactly as much ice on the air side of Γ as it removes on the ice side.

For the arithmetic law, $K_t = K_n = K_a + (K_i - K_a)\phi$ is affine, so $\Sigma_t = 0$. But $1/K_n$ is not affine, and Σ_n must be computed.

Step 3: computing Σ_n . Write $a = K_a$, $b = K_i$, $c = b - a$, and $K(p) = a + cp$. Substitute $p = \ell(u)$. Since $\ell' = \ell(1 - \ell)$, $du = dp/[p(1 - p)]$, and $u \gtrless 0$ becomes $p \gtrless \frac{1}{2}$:

$$\frac{\Sigma_n}{\varepsilon} = \int_0^{1/2} \left[\frac{1}{K} - \frac{1}{a} \right] \frac{dp}{p(1-p)} + \int_{1/2}^1 \left[\frac{1}{K} - \frac{1}{b} \right] \frac{dp}{p(1-p)} = -\frac{c}{a} \int_0^{1/2} \frac{dp}{K(1-p)} + \frac{c}{b} \int_{1/2}^1 \frac{dp}{Kp},$$

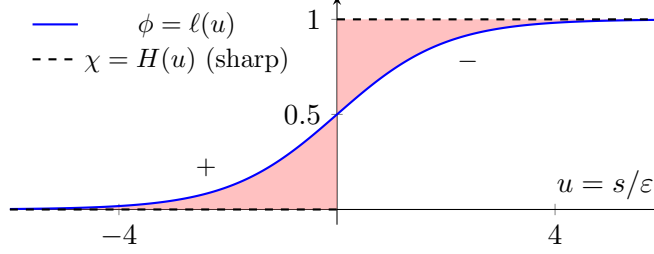


Figure 1: The tanh profile against the sharp indicator. The two shaded regions, $\phi - \chi$ on either side of the interface, are mirror images with opposite sign, so any quantity affine in ϕ has zero excess.

using $\frac{1}{K} - \frac{1}{a} = -\frac{cp}{aK}$ and $\frac{1}{K} - \frac{1}{b} = \frac{c(1-p)}{bK}$. Partial fractions, $\frac{1}{K(1-p)} = \frac{1}{b} \left[\frac{c}{K} + \frac{1}{1-p} \right]$ and $\frac{1}{Kp} = \frac{1}{a} \left[\frac{1}{p} - \frac{c}{K} \right]$, reduce both integrals to logarithms. With $K_m = K(\frac{1}{2})$, the $\ln 2$ terms cancel:

$$\frac{\Sigma_n}{\varepsilon} = \frac{c}{ab} \left[-\ln \frac{K_m}{a} - \ln \frac{b}{K_m} \right] \implies \Sigma_n^{\text{arith}} = -\varepsilon \left(\frac{1}{K_a} - \frac{1}{K_i} \right) \ln \frac{K_i}{K_a}. \quad (14)$$

At $K_i/K_a = 114.5$ the constant is $234.96 \text{ m K W}^{-1}$. The excess is negative because $1/K$ is convex in ϕ : within the transition the arithmetic law underestimates resistivity, so the transition offers too little resistance to heat crossing Γ . It acts as a thermal short across every interface.

No scalar law can satisfy both conditions. By Step 2, $\Sigma_t = 0$ calls for K affine and $\Sigma_n = 0$ for $1/K$ affine. Only a constant is both. The harmonic law $1/K = \phi/K_i + (1-\phi)/K_a$ fixes the normal condition and breaks the tangential one: $\Sigma_t^{\text{harm}} = -\varepsilon(K_i - K_a) \ln(K_i/K_a)$, by the same calculation with K and $1/K$ exchanged. Both values were checked against direct quadrature.

6 The tensor law

Proposition 1 asks for two separate things. Along the interface, $\mathbf{E}_t$ is fixed and the transition carries flux $K_t \mathbf{E}_t$, so its total conductance must match the sharp one: K_t must average correctly. Across the interface, q_n is fixed and the temperature drop is $q_n \int ds/K_n$, so its total resistance must match the sharp one: $1/K_n$ must average correctly. The general coefficient (8) lets us meet each requirement with its own function. By Step 2, the simplest choices are K_t affine and $1/K_n$ affine:

$$\mathbf{K}(\phi, \nabla \phi) = \underbrace{[K_a + (K_i - K_a)\phi]}_{K_t} (\mathbf{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}) + \underbrace{\left[\frac{\phi}{K_i} + \frac{1-\phi}{K_a} \right]^{-1}}_{K_n} \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}, \quad \hat{\mathbf{n}} = \frac{\nabla \phi}{|\nabla \phi|}. \quad (15)$$

Then $\Sigma_t = \Sigma_n = 0$ identically, the effective interface conditions (11)–(12) reduce to the sharp ones, and $\mathbf{K}_{\text{eff}}^\varepsilon - \mathbf{K}_{\text{eff}}^\star = O(\varepsilon^2)$.

Why this is the right fix.

1. *It targets the actual error.* Proposition 1 shows the first-order error consists of exactly two terms, and (15) sets each of them to zero. It is not a fit to any geometry.
2. *It needs direction dependence, and only that.* No scalar law can satisfy both conditions (§5). The tensor form is the minimal extension: it keeps the standard arithmetic law along the interface and changes only the response to normal gradients.

3. *It is robust to the profile.* The zero excesses rely only on the antisymmetry $\ell(-u) = 1 - \ell(u)$, not on the particular tanh shape. More generally, $\Sigma_t = 0$ holds for any K_t with $K_t(p) + K_t(1-p) = K_a + K_i$, and $\Sigma_n = 0$ for any K_n with $1/K_n(p) + 1/K_n(1-p) = 1/K_a + 1/K_i$. The affine choices are the simplest members of these families.
4. *It leaves the sharp problem untouched.* Both branches take the pure-phase values at $\phi \in \{0, 1\}$, so $\mathbf{K} = K_\star \mathbf{I}$ outside the transition and Lemma 1 still holds. The sharp limit is unchanged.
5. *It is well posed with the same bounds.* $\mathbf{K}$ is symmetric with eigenvalues K_t (along) and K_n (across), both in $[K_a, K_i]$. The cell problem stays uniformly elliptic, $\mathbf{K}_{\text{eff}}^\varepsilon$ is symmetric positive definite, and the discrete operator stays symmetric positive definite.
6. *It has no free parameter* and uses only ϕ and $\nabla\phi$, which the phase field already provides. Where $\nabla\phi = 0$, $\hat{\mathbf{n}}$ is undefined, but that happens only in the pure phases, where $K_t = K_n$ and $\mathbf{K}$ is isotropic whatever $\hat{\mathbf{n}}$ is.

7 Size of the arithmetic-law bias

Let $T^\star$ be the sharp solution for a unit mean gradient $\mathbf{e}$, and $\mathbf{E}_t$, q_n its tangential gradient and normal flux on Γ . The first-order correction $\tau = T^\varepsilon - T^\star$ solves the sharp bulk equations with jumps (11)–(12). Two applications of the divergence theorem, one against $T^\star$ and one against its corrector, plus the flux $\Sigma_t \mathbf{E}_t$ carried along Γ , give

$$\mathbf{e} \cdot \mathbf{K}_{\text{eff}}^\varepsilon \mathbf{e} - \mathbf{e} \cdot \mathbf{K}_{\text{eff}}^\star \mathbf{e} = \frac{1}{|\Omega|} \int_\Gamma \left(\Sigma_t |\mathbf{E}_t|^2 - \Sigma_n q_n^2 \right) d\Gamma + O(\varepsilon^2). \quad (16)$$

For the arithmetic law only the second term survives, and since $\Sigma_n < 0$ the bias is positive. The cell conducts too well wherever heat crosses an interface, in proportion to ε , to the interfacial area, and to the square of the normal flux.

Planar slab: the exact diffuse solution. Take an ice slab with two flat faces per period L , and ϕ a function of y only. The cell problem is then one-dimensional and solvable exactly for any ε . Along the faces, $t = 0$ and $k_\parallel = \langle K_t \rangle$. Across them, $K_n(t' + 1)$ is constant and $\langle t' \rangle = 0$, so $k_\perp = \langle 1/K_n \rangle^{-1}$. For the arithmetic law, Step 2 gives $\langle K_t \rangle$ equal to its sharp value at every ε , while

$$\langle 1/K_n \rangle = \frac{\bar{\phi}}{K_i} + \frac{1 - \bar{\phi}}{K_a} + \frac{2 \Sigma_n}{L} \quad (17)$$

up to exponentially small overlap between the two faces' tails. This is exact in resistivity, and (16) is its linearization in ε . For the tensor law, $1/K_n$ is affine and $k_\perp$ is exact at every ε . At $\bar{\phi} = \frac{1}{2}$:

ε	arithmetic $k_\perp$	exact $k_\perp$	bias
$L/50$	0.0632	0.03965	+59.4%
$L/128$	0.0464	0.03965	+17.0%
$L/512$	0.0412	0.03965	+3.8%

The 59% means that the arithmetic diffuse cell reports a conductivity across the slab of $0.063 \text{ W m}^{-1} \text{ K}^{-1}$ for a geometry whose true value is 0.040. The microstructure is the same; only the numerical width of its interfaces differs.

8 Verification

Two cells with closed-form sharp solutions, run over an ε ladder at fixed $\varepsilon/h = 5.12$, so the interface error is not confounded with mesh convergence.

Planar slab ($\bar{\phi} = \frac{1}{2}$). The normal is a fixed axis, so this case tests the normal branch of (15) in isolation.

ε	arithmetic (exact, (17))	tensor (measured)
$L/50$	+59.4%	$< 4 \times 10^{-7}$
$L/64$	+41.1%	$< 4 \times 10^{-7}$
$L/128$	+17.0%	$< 4 \times 10^{-7}$
$L/256$	+7.9%	$< 4 \times 10^{-7}$
$L/512$	+3.8%	$< 4 \times 10^{-7}$

The tensor law reproduces the exact $k_{\perp}$ to solver precision at every rung, and $k_{\parallel}$ to 8×10^{-7} . The fitted slope of $1/k_{\perp}$ against ε is 10^{-6} of the arithmetic law's.

Disk array (square array, $f = \pi R^2/L^2 = 0.196$). Here the normal takes every direction, so the full tensor, including its off-diagonal terms, is exercised. The sharp value is Rayleigh's, $k = 0.0295684 \text{ W m}^{-1} \text{ K}^{-1}$ (Rayleigh 1892; Perrins et al. 1979). Putting the sharp dipole field into (16) predicts the arithmetic law's first-order slope with no fitted constant,

$$\frac{dk}{d\varepsilon} = \frac{|\Sigma_n|}{\varepsilon} \frac{\pi R}{L^2} \left[\frac{K_a(1 + \beta)}{1 - \beta f} \right]^2, \quad \beta = \frac{K_i - K_a}{K_i + K_a}. \quad (18)$$

ε	arith. (pred.)	arith. (meas.)	tensor (meas.)	tensor order
$L/100$	+15.07%	+18.19%	+1.212%	
$L/200$	+7.53%	+8.26%	+0.204%	2.57
$L/400$	+3.77%	+3.95%	+0.034%	2.59
$L/800$	+1.88%	+1.93%	+0.006%	2.50

The arithmetic law's fitted first-order slope is within 1.9% of (18), which confirms (16) on a curved interface. The tensor law's error is at most $0.067\times$ the arithmetic error and falls at an observed order of about 2.5, so no first-order term remains. The measured ice fraction equals the exact diffuse-disk value $f(1 + \pi^2\varepsilon^2/3R^2)$ to 3×10^{-10} , so the errors come from the conductivity law, not from the initial condition. Drivers: `enceladus_DSM/studies/keff_sharp_limit/{verification,disk}/`.

9 Implementation

The cell problem (7) is solved with isogeometric Galerkin discretisation (PetIGA) in `enceladus_dsm`, in-line with `-keff 1` or on saved snapshots with `-keff_replay`. The law is selected with `-keff_interp tensor` (the default) or `arith`. At each quadrature point, `KeffPointCond` builds $\mathbf{K} = K_t \mathbf{I} + (K_n - K_t) \nabla \phi \otimes \nabla \phi / |\nabla \phi|^2$ from the projected ϕ and $\nabla \phi$. The stiffness $\int \nabla N_a \cdot \mathbf{K} \nabla N_b$ is assembled once and shared by all directions. The load is $-\int \nabla N_a \cdot \mathbf{K} \mathbf{e}_m$, and $\mathbf{K}_{\text{eff}}$ is the quadrature average of $\mathbf{K}(\nabla t_m + \mathbf{e}_m)$.

10 Scope and limitations

- *Steady state only.* Everything here concerns the homogenization cell problem on a frozen ϕ . It says nothing about, and is not an argument for changing, the transient equations of the phase-field model that produced ϕ .
- *Residual $O(\varepsilon^2)$.* This includes curvature terms and a geometric one: a diffuse disk holds $\pi^3\varepsilon^2/3$ more area of ice than the sharp disk. The observed rate on the disk is about 2.5. We have not derived the second-order coefficient, so the rate is an observation.
- *Thin necks.* The inner analysis assumes the transition is thin compared with the curvature radius and with the distance to other parts of the interface. Where two interfaces come within a few ε of each other, as in young sinter necks, the transitions overlap and no interpolation law is justified by it. This limits how well the phase field resolves the geometry, and it applies to the phase-field geometry as much as to $\mathbf{K}_{\text{eff}}$.
- *Profile.* The argument uses the antisymmetry of the profile about $\phi = \frac{1}{2}$. A profile distorted by the phase-field evolution changes the excesses only in proportion to its departure from antisymmetry.
- *Geometry error is untouched.* The law removes the error of homogenizing a given ϕ . It does not correct errors in ϕ itself.

11 Related work

- The expansion of §4 is the standard thin-interface analysis of phase-field models (Karma & Rappel 1998). The first-order interface corrections that arise with unequal conductivities in the two phases are analysed by Almgren (1999) and McFadden, Wheeler & Anderson (2000).
- Elliptic problems posed on a smoothed indicator are the setting of the diffuse-domain method. Li et al. (2009) introduce it, and Lervåg & Lowengrub (2015) use matched asymptotics to show which formulations are first- or second-order accurate in ε .
- In phase-field elasticity, Durga, Wollants & Moelans (2013) show that interpolating the stiffness creates interfacial excess contributions. Schneider et al. (2015) remove them by interpolating so that the mechanical jump conditions (continuous normal traction and tangential strain) hold inside the diffuse interface. That is the elastic counterpart of keeping q_n and $\mathbf{E}_t$ continuous here.

12 Suggested Methods text

The effective conductivity tensor is obtained by periodic homogenization (Calonne et al., 2011). The two-domain cell problem is written as a single weak equation with the piecewise conductivity $K_\star = K(\chi)$ of the ice indicator χ , and solved on the phase field by replacing χ with ϕ . Inside the diffuse interface the conductivity must be interpolated. A matched asymptotic expansion shows that, to first order in the interface width ε , the diffuse interface acts as a sharp one with a spurious thermal resistance Σ_n across it and a spurious conductance Σ_t along it, both computable in closed form for the tanh profile. The arithmetic interpolation gives $\Sigma_t = 0$ but $\Sigma_n = -\varepsilon(K_a^{-1} - K_i^{-1}) \ln(K_i/K_a)$, which overestimates conduction across interfaces by up to tens of percent at $K_i/K_a \approx 115$. We therefore use arithmetic interpolation for gradients along the interface and harmonic interpolation for gradients across it, $\mathbf{K} = K_{\text{ar}}(\phi)(\mathbf{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}) + K_{\text{h}}(\phi) \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}$

with $\hat{\mathbf{n}} = \nabla\phi/|\nabla\phi|$. This makes both terms vanish, so the error is $O(\varepsilon^2)$. The law reduces to the pure-phase conductivities outside the interface and has no adjustable parameters. We verified it against closed-form sharp solutions for a planar slab and a square array of disks (Rayleigh, 1892; Perrins et al., 1979).

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